GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Instrument Cable-BBOU/ELEP/AKAB
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Specification for Instrument Cable-BBOU/ELEP/AKAB

R02	05.08.22	Re-Issued for Review	P. Roy	P. Olukah	A. Duru	
R01	18.07.22	Issued for Review	P. Roy	P. Olukah	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	18.07.22	Issued for Review
R02	05.08.22	Re-issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-040

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

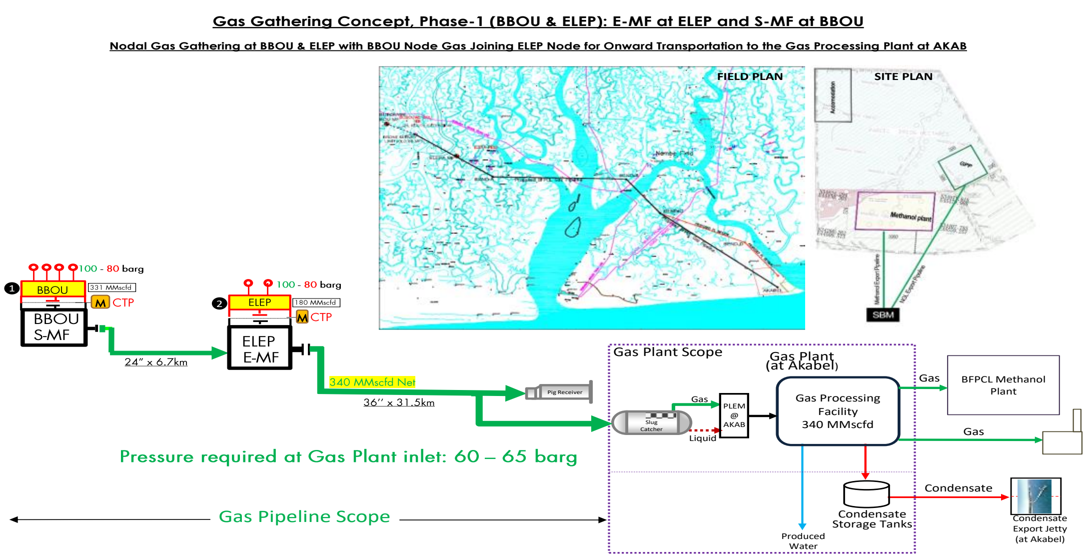

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document defines the basic requirements for the engineering and design of Instrumentation cables for BBOU, ELEP and AKAB pipeline.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Instrument Cable and Wiring

3.1. Cable

3.1.1. Standard instrumentation cable shall be used in accordance with IEC standards, Shell standards and other international standards. The insulation voltage level shall be 300/500V for signal cables and 600/1000V for low voltage power cables

3.1.2. All cable shall meet the requirement of IEC-60331 and IEC-60332 fire retardant and fire resistant characteristics.

3.1.3. Main cable runs between control room and field junction box shall be multipair, twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, overall screen and with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed. Cables between Instruments and junction boxes shall be 1.5 mm2. All the multi pair/multi triad cables shall be have both individual pair screen and overall shield for both digital and analog signal cables. The conductors of power supply cables shall have a minimum cross sectional area of 2.5 mm2.

3.1.4. Primary Insulation shall be XLPE for all signal cables. Primary insulation for XLPE insulated cables, shall be of 90°C as per EN 50290-2-29 Thickness of primary insulation shall be 0.5 mm as a minimum

3.1.5. The thickness of the sheath shall be as per EN 50290-2-22

3.1.6. Inner and outer sheath of cable shall be flame retardant and low smoke made of extruded PVC Type TM 53 (90°C) as per EN 50290-2-22 and shall meet the following requirements

a. Minimum Oxygen index of PVC shall be 30 at 27 °C ±2°C.
b. Temperature index shall be over 250 °C.
c. Inner and Outer sheath shall meet flame retardant requirements for bunched cables as per IEC 60332-3-22 category A.
d. A rip cord shall be provided for inner sheath.
e. Outer sheath shall be suitable for protecting the cable against rodent and termite attack.
f. Outer sheath shall be UV resistant.

3.1.7. Sequential marking of the length of the cable in meters shall be provided on the outer sheath at every one meter. The embossing /engraving/printing shall be legible and indelible.

3.1.8. Tolerance in overall diameter of cable shall be within ± 2 mm over offered value.

3.1.9. Tolerance over the total ordered length for a type of cable shall be as follows:

a. ± 5% for total length less than 5 km.
b. ± 2% for total length more than 5 km.

3.1.10. Specific Requirements for Fire Resistant Cable (Circuit Integrity Cable).

- The cables shall have circuit integrity as per IEC 60331-21.
- Primary insulation shall be heat resisting elastomeric which can withstand temperature up to 90°C such as silicon rubber/mica glass tape/EPR (medium grade) as per EN 50290-2-36. Insulation thickness shall be 0.5 mm minimum confirm to EN-50288-7.
- A wrapping of tape made of PETP (polyethylene "Terephthalate)/woven glass shall be provided over core insulation for mica glass tape. This is not applicable for silicon rubber; minimum thickness shall be 1.0 mm for mica application as per EN-50288-7
- Each multi-pair cable shall be provided with overall shield and individual pair/triad shield to reduce or prevent possible crosstalk from adjacent line circuit or externally induced interference from outside sources such as electromagnetic effects of lightning currents. Overall shield shall comply with the following requirements:
  - ✓ Overall shield shall be a non-hygroscopic aluminium-polyester tape, helically applied to provide 100% coverage. Aluminium shall be on the outside in continuous contact with the overall drain wire.
  - ✓ Overall drain wire shall be stranded, tinned copper conductor.
  - ✓ Grounding of the shield shall be done at one point only, located inside marshalling cabinets of each technical room, on instrument side, shield shall be isolated by means of a sleeve.

3.1.11. Inner and outer sheath shall be made of low smoke, heat resistant, oil resistant and flame-retardant material with oxygen index over 30, temperature index shall be over 250°C. Acid generation shall be maximum 20% by weight as per IEC 60754. Smoke density rating not to exceed 60% as per ASTM D 2843.

3.1.12. The thickness of the sheath shall be as per EN 50290-2-22. Inner and outer sheath colour shall be Red. A rip cord shall be provided for inner sheath.

3.1.13. Insulation over inner sheath shall be of special high oxygen index, low smoke halogen free fire resisting compound.

3.1.14. Cable Armour shall be galvanized iron wire.

3.2. Electrical Characteristics

3.2.1. Maximum DC resistance of the conductor of the finished cable shall not exceed 18.1 Ω / km at 20°C for cables with 1.0 mm2 conductors, 12.1 Ω / km at 20°C for cables with 1.5 mm2 conductors and 7.41 Ω / km at 20°C for cables with 2.5 mm2 conductors as per EN 60228. The insulation grade shall be 500 V as a minimum and shall meet insulation resistance, voltage and spark test requirements as per EN-50288-7.

3.3. Capacitance:

3.3.1. Mutual Capacitance for PVC Insulated cables the mutual capacitance of adjacent cores/of a pair/or with respect to side of a quad shall not exceed of 250 pF / metre at a frequency of 1 KHz.

3.3.2. Mutual Capacitance for XLPE Insulated cables the mutual capacitance between adjacent cores / of a pair (or with respect to the side of a quad) shall not exceed of 90 pF / metre at a frequency of 1 KHz.

3.3.3. Unbalanced Capacitance between pairs/triads/Quads The capacitance between any pairs/triads/Quads shall not exceed a maximum of 400 pF / metre at a frequency of 1 KHz.

3.3.4. L/R ratio of adjacent core shall not exceed 40 μH / Ω for cables with 1.5 mm2 conductors and 25 micro h / Ω for cables with 0.5 mm2 conductors as per EN-50289-1-12

3.3.5. The drain wire resistance including shield shall not exceed 30 Ω / km.

3.3.6. Electrostatic noise rejection ratio of the finished cable shall be over 76 dB.

3.3.7. Outer jacket colour will be:

- Black for Non-IS PLC cable
- Light Blue for IS PLC cable
- Light Blue with Gray stripes for IS ESD cable
- Red for Non-IS ESD/F&G cable

- Gray for 24VDC Non-IS Solenoid valve (ESD)
- Black for 24VDC Non-IS Solenoid valve (PLC)

3.3.8. Wiring

- Both ends of all cables and their used cores shall be identified by individual cable marker rings as per drawing.
- To achieve insulation and continuity of the screen, the drain wires shall be accommodated using terminals with in the box.
- Wire colour code shall follow Shell DEP standard.
- Only continuous conductors shall be used. No splicing shall be permitted. Marking by inscription for identification on each core and sheath shall be legible and permanent.
- Normally the cores shall be 6,6 pairs/triads.
- About 20% spare core / pairs shall be provided in all multicore / multi pair cables.

4. Inspection and Packing

4.1. Cables shall be dispatched in wooden drums securely battened with take-off end fully protected against damage and ends of cables shall be sealed by means of non-hygroscopic material. On all cable drums, following information shall be stencilled on outside of the flanges,

- Name of manufacturer and client
- Purchase order number
- Drum number
- Type of cable and cable length
- Gross weight
- Direction of rotation of drum

4.2. All cables shall be clearly and permanently marked, the distance between the two markings shall not exceed 1.5m, on their external sheath with the following data as a minimum:

- Conductor size
- Number of twisted pairs
- Voltage Rating
- Type of insulation
- Temperature rating
- Manufacturer's name
- Standard marking
- Fire resistance rating
- Cable Type
- Manufacturing Date
- Length Marking

4.3. All cabling system shall be tested and inspected being witnessed by the purchaser or his representative. The tests and Inspection shall be approved in the visual check of workmanship finish and quantities, measuring of dimensions, measurement of insulation resistance, operation test. Manufacture test (FAT), Site acceptance test (SAT) Cables shall be factory tested by the manufacturer to ensure that they have been manufactured in accordance with this specification and the applicable standards.

4.4. Buyer reserves the right to inspect and test either the whole length of the cable drum/reel/coil or on a sample basis at the manufacturer's plant before shipment and/or in the field.

5. Vendor Responsibities

5.1. Unit prices shall remain same till the completion of the order.

5.2. Unless otherwise indicated in the specifications, drum length for each type shall be 1000meter except the last drum.

5.3. Drum length tolerance shall not exceed +2%, -0%.

5.4. Outer sheath of cable shall be embossed with,

- Voltage grade
- Manufacturer identification
- Number of core & conductor size
- Drum progressive length at every meter

5.5. Vendor shall provide the test certificates for all the tests.

5.6. Vendor shall furnish following electrical data for each type of cable

- Conductor resistance per km at 20° C
- Insulation resistance
- Mutual capacitance & inductance per km
- L/R ratio
- Oxygen & temperature index

5.7. Vendor shall furnish following construction details for each type of cable

- Conductor diameter
- Insulation thickness
- Number of Twists per meter, if applicable
- Screen and drain wire details, if applicable
- Inner sheath thickness
- Outer sheath thickness
- Cable gross weight

5.8. Vendor shall furnish following dimensional details for each type of cable

- Diameter over inner sheath
- Diameter under armour
- Diameter over armour
- Diameter of armour

- Overall diameter
-

APPENDIX – A

Standards	**Description**
D 2843	Standard Test method for Density of smoke from the Burning or Decomposition of Plastic
D 2863	Test method for measuring the minimum oxygen concentration to support candle like combustion of plastics (oxygen index)
BS-10244-2	Steel Wire and Wire Products-Non-ferrous metallic coatings on steel wire, Part 2: Zinc or Zinc Alloy Coatings
BS-10257-1	Zinc or Zinc Alloy coated non-alloy steel wire for Armouring either power cables or telecommunication Cables, Part 1: Land Cable
BS-50288	Multi-element metallic cables used in analogue and digital communication and control Part 1 General Specification. Part- 7 Sectional specification for instrumentation and control cables
BS-50289-1	Communication cables. Specifications for test methods. Electrical test methods. Capacitance. SEC 5 Capacitance SEC 12 Inductance
BS-50290-2	Communication cables. Common design rules and construction- 22 PVC Sheathing Compound 29 Cross-linked PE insulation compounds 36 Common design rules and construction - Cross Linked Silicone rubber insulation compound
BS-50363-1	Insulating, sheathing and covering materials for low voltage Energy cables. Cross-linked elastomeric insulating Compounds.
BS-60228	Conductors for Insulated Cables
BS-60754-1	Test on gases evolved during combustion of materials from cables. Determination of the halogen acid gas content. IEC International Electro technical Commission.
IEC-60331	Testing of Fire Resistant cables. Part-21 Procedures and requirements - Cables of rated voltage up to and including 0,6/1,0 kV
IEC-60332	Tests on bunched wires and cables
IEC-60332-3-22	Test for vertical flame spread of vertically-mounted bunched wires or cables – Category A
IEC-61034	Measurement of smoke density of cables burning under defined conditions Part-2 Test Procedure and requirements.
IEC-60754	Test on Gases Evolved during Combustion of Materials from Cables – Part 1, 2
IEC-60811	Test methods for insulation and sheaths of electric Cables
UL 1581 article 1200	Reference Standard for Electrical Wires, Cables, and Flexible Cords